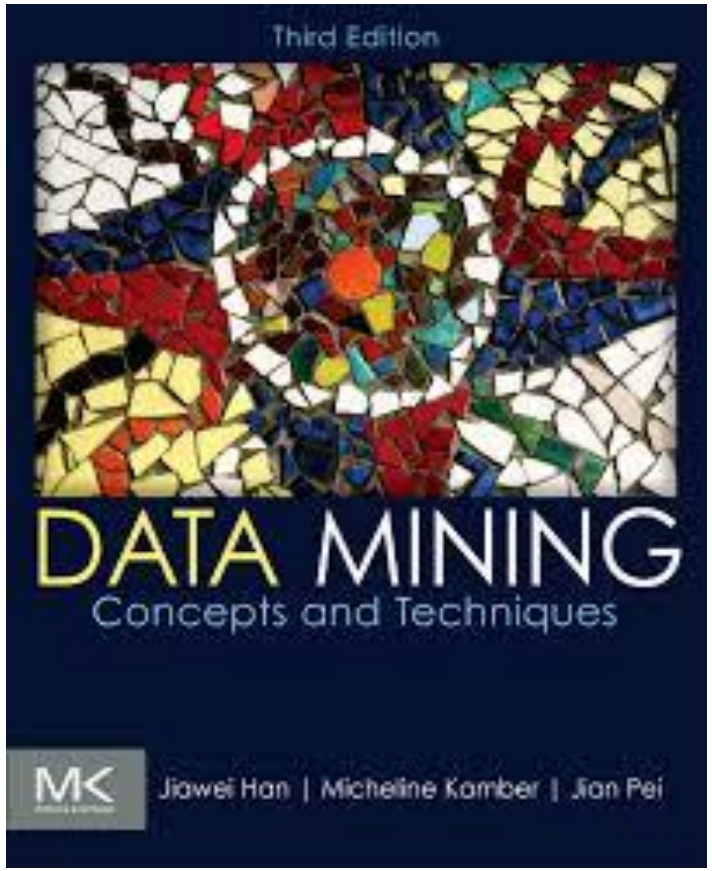
Machine Learning and Data Mining Basics

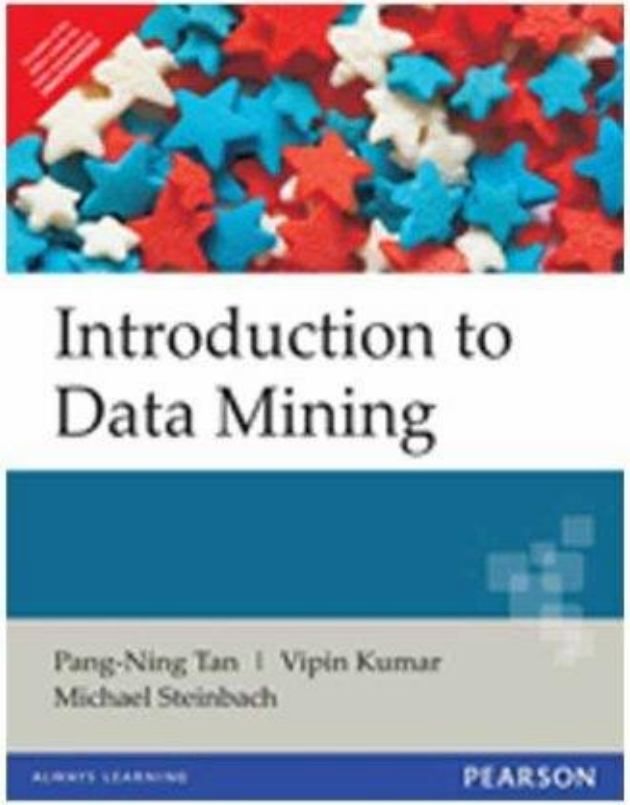
Course Goal

- **Differentiate Data Mining and Machine Learning** – Understand the scope, purpose, and key differences.
- **Understand Data Representation** – Learn how data is structured, formatted, and encoded for ML.
- **Master Data Preprocessing** – Handle missing data, scaling, normalization, and feature engineering.
- **Comprehend the Machine Learning Life Cycle** – From problem definition to model deployment.
- **Identify Types of Machine Learning** – Supervised, Semi-supervised, Unsupervised, and Reinforcement Learning.
- **Implement Core ML Algorithms** – Linear & logistic regression, decision trees, SVM, KNN, Naïve Bayes, etc.
- **Evaluate ML Models** – Metrics, validation, cross-validation, and performance improvement techniques.
- **Introduction to Deep Learning** – Basics of neural networks and their applications.
- **Develop Practical Thinking Skills** – Learn frameworks for problem-solving and common beginner pitfalls.

Course Materials and References

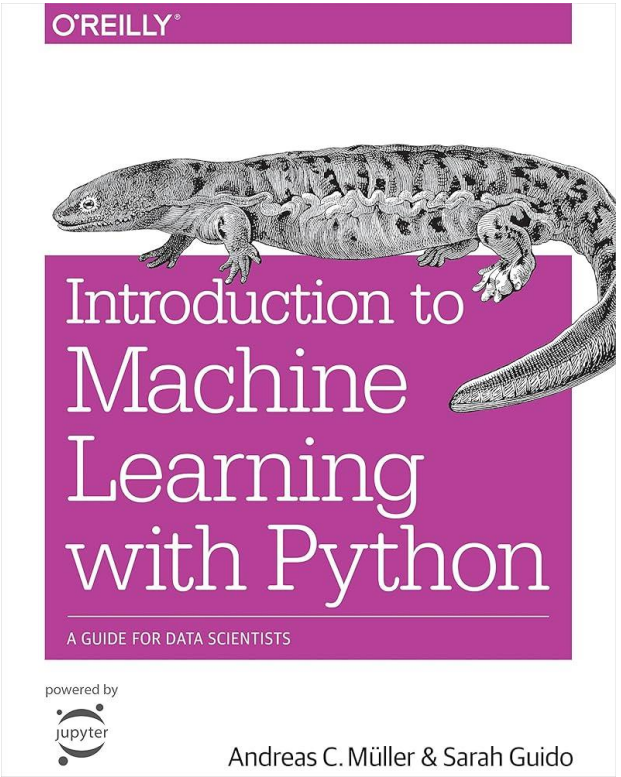


Data Mining Concepts and Techniques by Jiawei Han, Micheline Kamber, Jian Pei ·

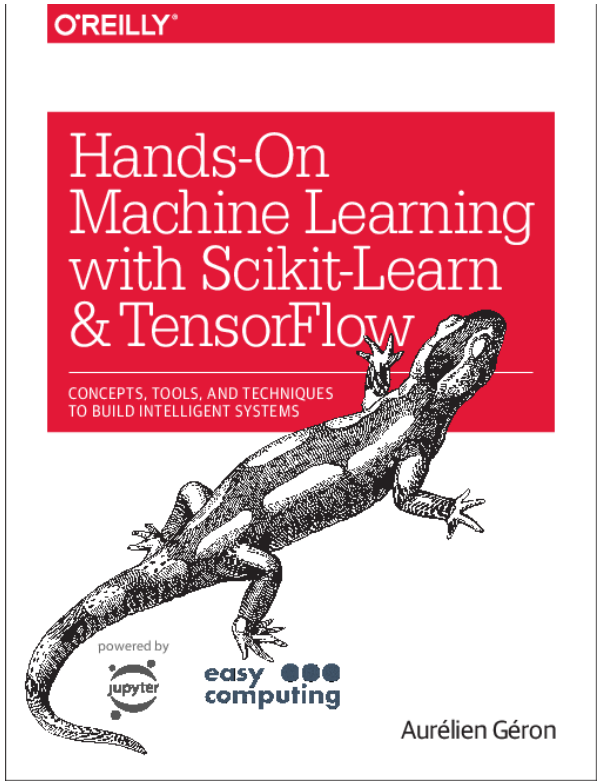


Introduction to Data Mining by Pang-Ning Tan Michael Steinbach Vipin Kumar

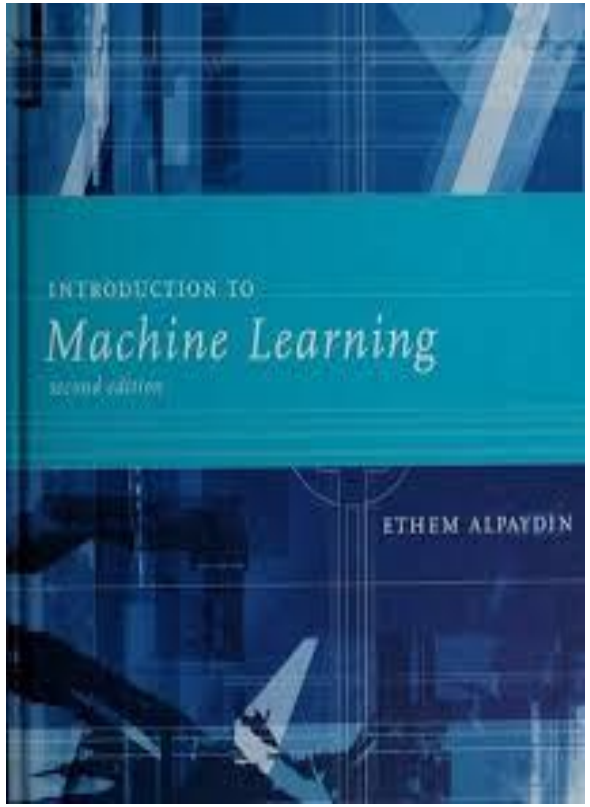
Course Materials and References



Introduction to Machine Learning with Python by Andreas C. Muller and Sarah Guido



Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurélien Géron



Introduction to Machine Learning by Ethem Alpaydın

Data Mining

- Fundamentals of Data Mining
  - What is Data Mining?
  - Knowledge discovery in databases (KDD)
  - Multi-Dimensional View of Data Mining
  - What Kinds of Data Can Be Mined?
  - What Kinds of Patterns Can Be Mined?
  - Issues in Data Mining
- Data Representation and Understanding
  - Data Objects and Attribute Types
  - Basic Statistical Descriptions of Data
  - Data Visualization

What is Data Mining?

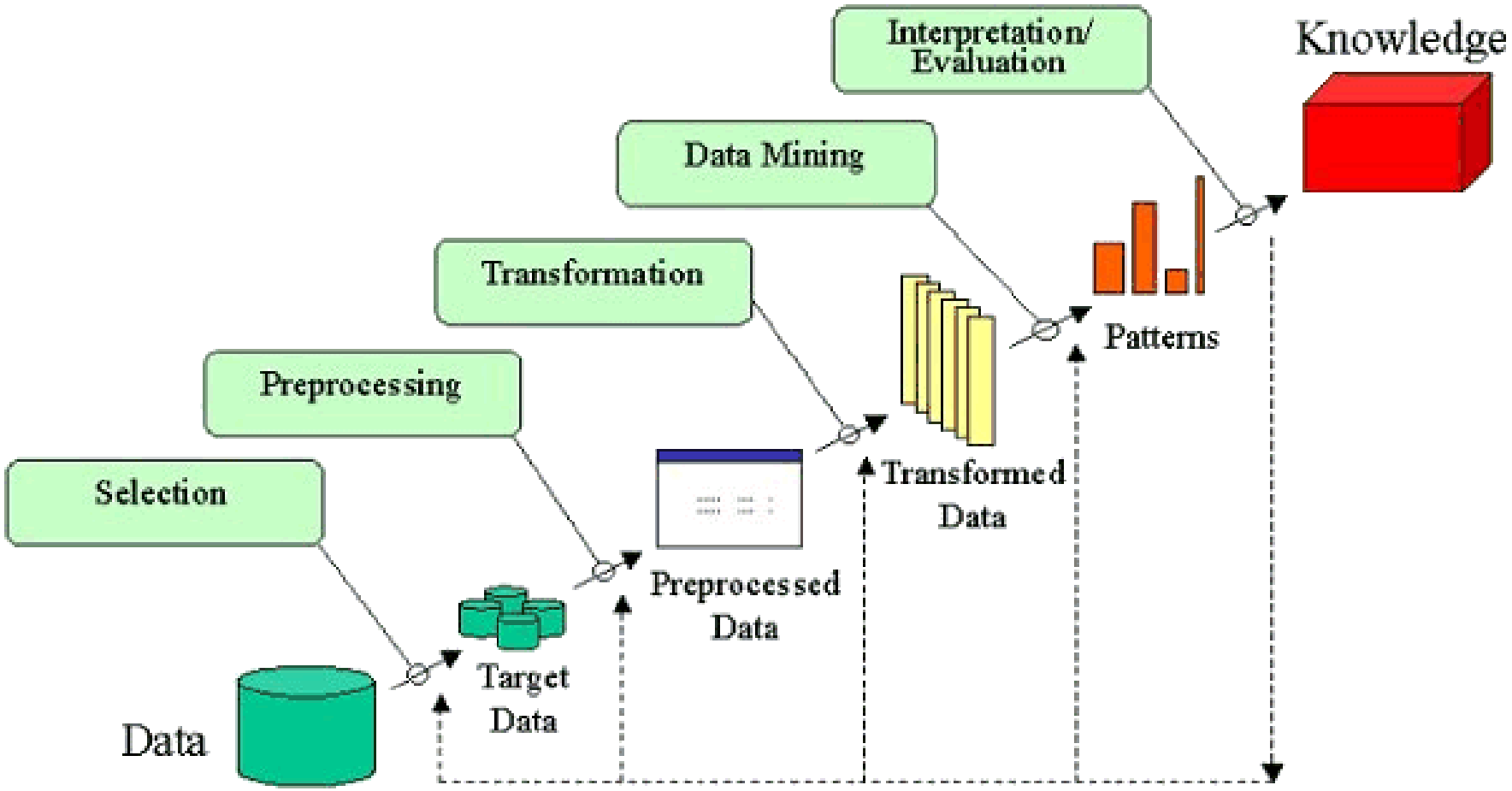
- Data Mining is the process of discovering patterns, relationships, anomalies, and useful information from large datasets.

- Also known as Knowledge discovery in databases (KDD) , data/pattern analysis, data archeology, data dredging, information harvesting , business intelligence.

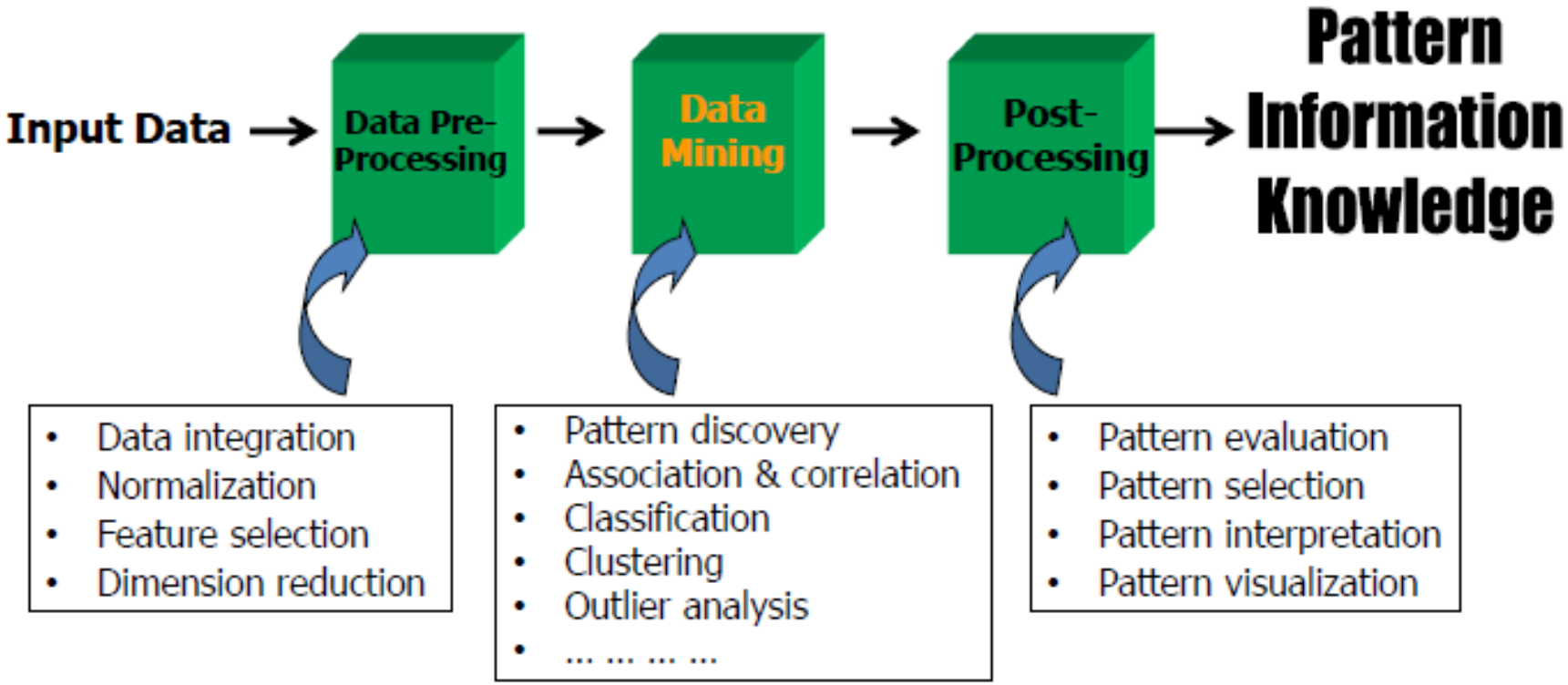
What is Data Mining?

- Broader than machine learning.
- Includes:
  - Statistical analysis
  - Database systems
  - Preprocessing
  - Visualization
  - Knowledge discovery.

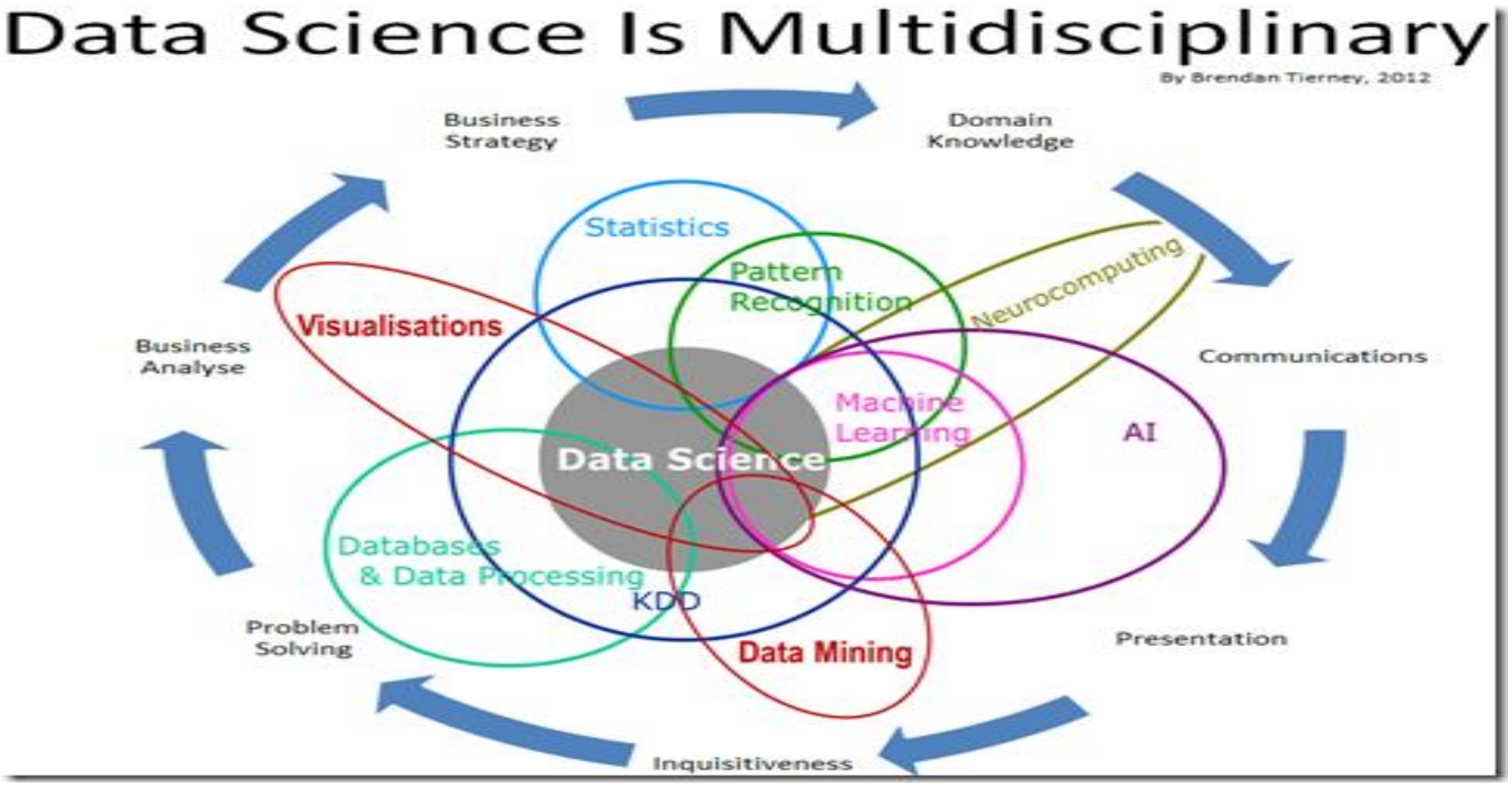
Knowledge Discovery (KDD) Process



KDD Process: A Typical View from ML and Statistics



Multi-Dimensional View of Data Mining



What Kinds of Data Can Be Mined?

Database-oriented data sets and applications:

- Relational database, data warehouse, transactional database.
- Object-relational databases, Heterogeneous databases and legacy databases.

Advanced data sets and advanced applications:

- Data streams and sensor data.
- Time-series data, temporal data, sequence data (such as: bio-sequences).
- Structure data, graphs, social networks and information networks.
- Spatial data and spatiotemporal data.
- Multimedia database.
- Text databases.
- The World-Wide Web.

What Kinds of Patterns Can Be Mined?

1. Generalization:

The process of transforming low-level, detailed data into higher-level summarized concepts.

Raw data → Knowledge

Raw Value	Generalized Level 1	Level 2
23 years	Young	Adult
45 years	Middle-aged	Adult
72 years	Senior	Elderly

City	State	Country
Boston	Massachusetts	USA

What Kinds of Patterns Can Be Mined?

❖ Real-world datasets are:

- Noisy
- Too detailed
- Hard to interpret
- Hard to model directly

❖ Generalization:

- Reduces complexity
- Makes patterns visible
- Improves interpretability
- Supports decision-making

Customer ID	Age	Income	Zip Code

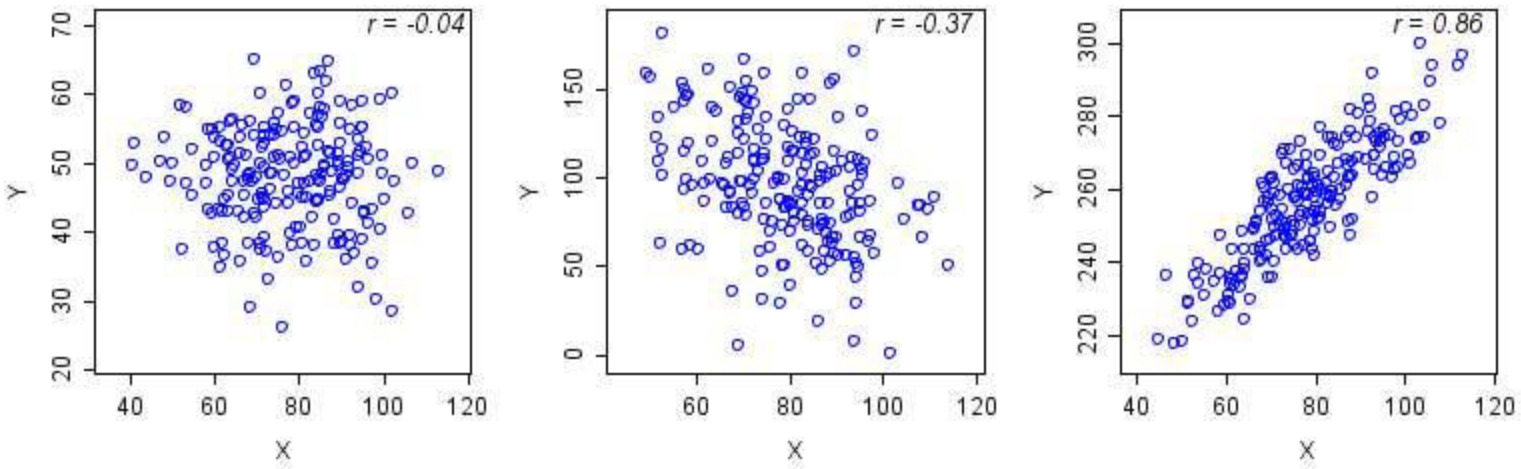
Age Group	Income Level	Region
Young	Medium	Northeast

What Kinds of Patterns Can Be Mined?

2. Association and Correlation Analysis

- Frequent patterns (or frequent item sets)

  What items are frequently purchased together in your Market?



What Kinds of Patterns Can Be Mined?

3. Classification and Label prediction (Supervised Learning)

- Construct models (functions) based on some training examples
  - Describe and distinguish classes or concepts for future prediction

    E.g., classifying countries based on (climate), or classifying cars based on (fuel consumption)
  - Predict some unknown class labels

- Typical Methods
  - Decision Trees, Naïve Bayesian Classification, Support vector machines, Neural networks, Rule-based classification, Pattern-based classification, Logistic regression

- Typical Applications:

Fraud detection, Classifying Diseases.

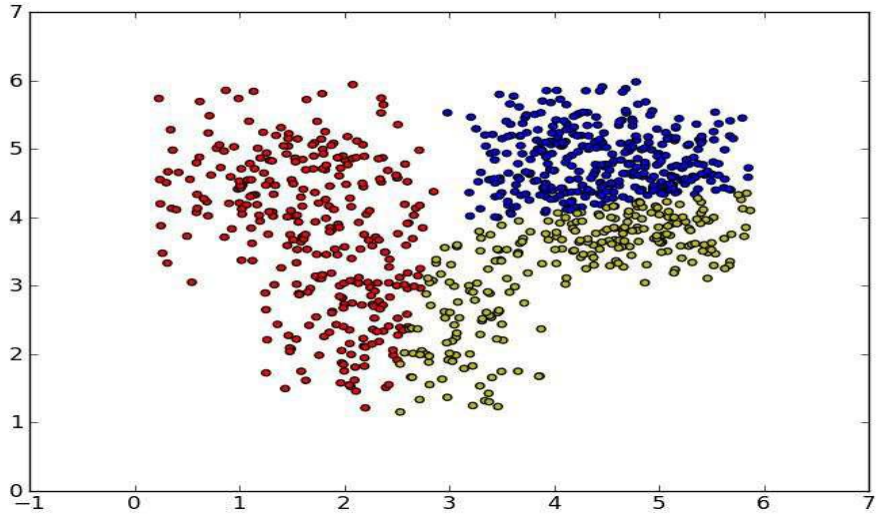
What Kinds of Patterns Can Be Mined?

4. Cluster Analysis (Unsupervised learning - Class label is unknown)

Group data to form new categories (i.e., clusters)

Principle:

Maximizing intra-class similarity & minimizing interclass similarity



What Kinds of Patterns Can Be Mined?

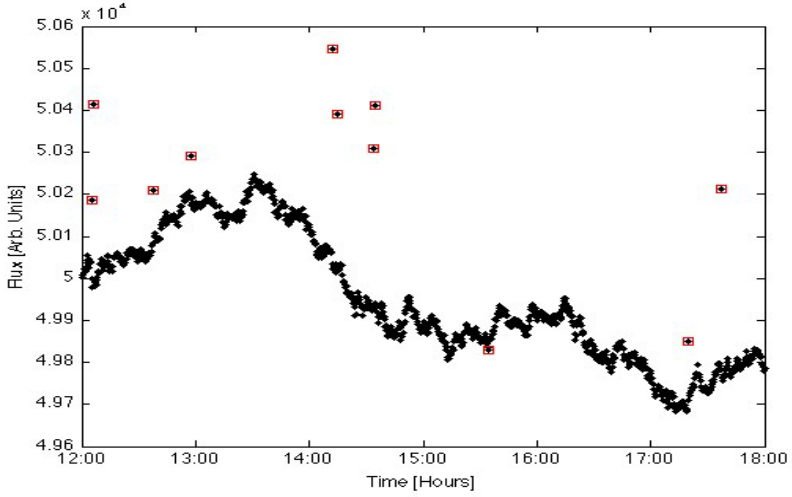
5. Outlier Analysis

A data object that does not comply with the general behavior of the data is called an **outlier**.

Methods:

By product of clustering or regression analysis

❖ Useful in Fraud Detection



Data Mining Issues

- Mining Methodology
  - Mining various and new kinds of knowledge.
  - knowledge in multi-dimensional space.
  - Handling noise, uncertainty, and incompleteness of data.
- User Interaction
  - Presentation and visualization of data mining results.
- Efficiency and Scalability
  - Efficiency and scalability of data mining algorithms
  - Parallel, distributed, and Stream methods

Data Mining Issues

- Diversity of data types
  - Handling complex types of data
  - Mining dynamic, networked, and global data repositories
- Data mining and society
  - Social impacts of data mining
  - Privacy-preserving data mining
  - Invisible data mining

Data Mining

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  -
- Data Representation and Understanding
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  - Basic Statistical Descriptions of Data
  - Data Visualization

Data Representation and Understanding

❖ What is Data?

➢ Collection of data objects and their attributes

- Data sets are made up of data objects
- A data object represents an entity

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Data Objects

Examples:

- Sales database: customers, store items, sales.
- Medical database: patients, treatments.
- University database: students, professors, courses.

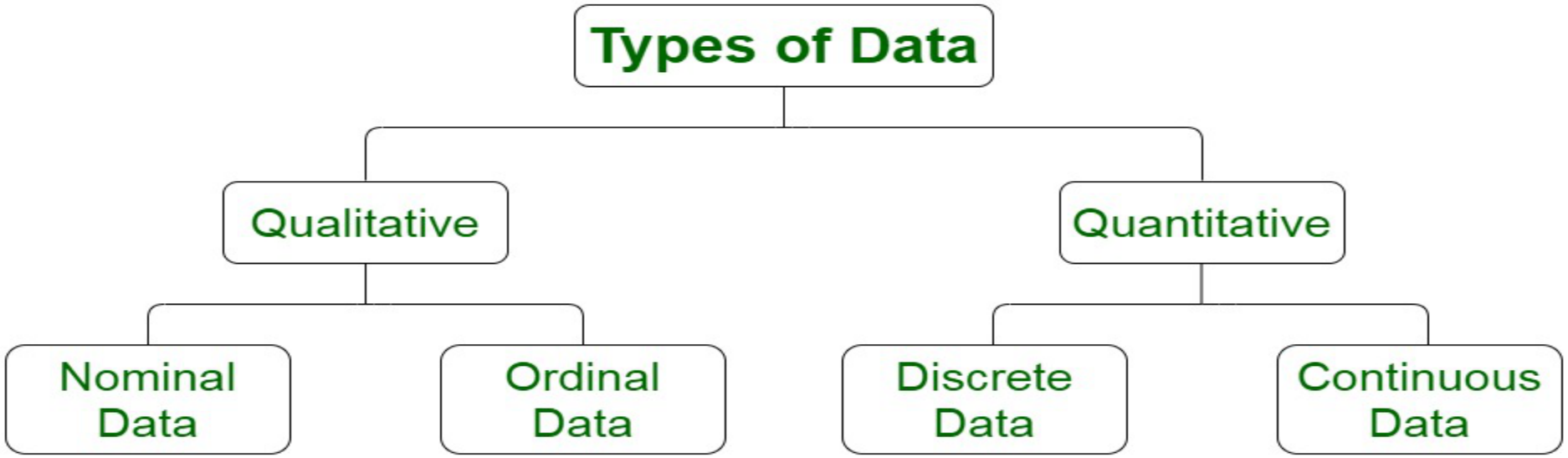
❖ Also called samples , examples, instances, data points, objects, tuples

❖ Data objects are described by attributes

Database rows → data objects; columns →attributes

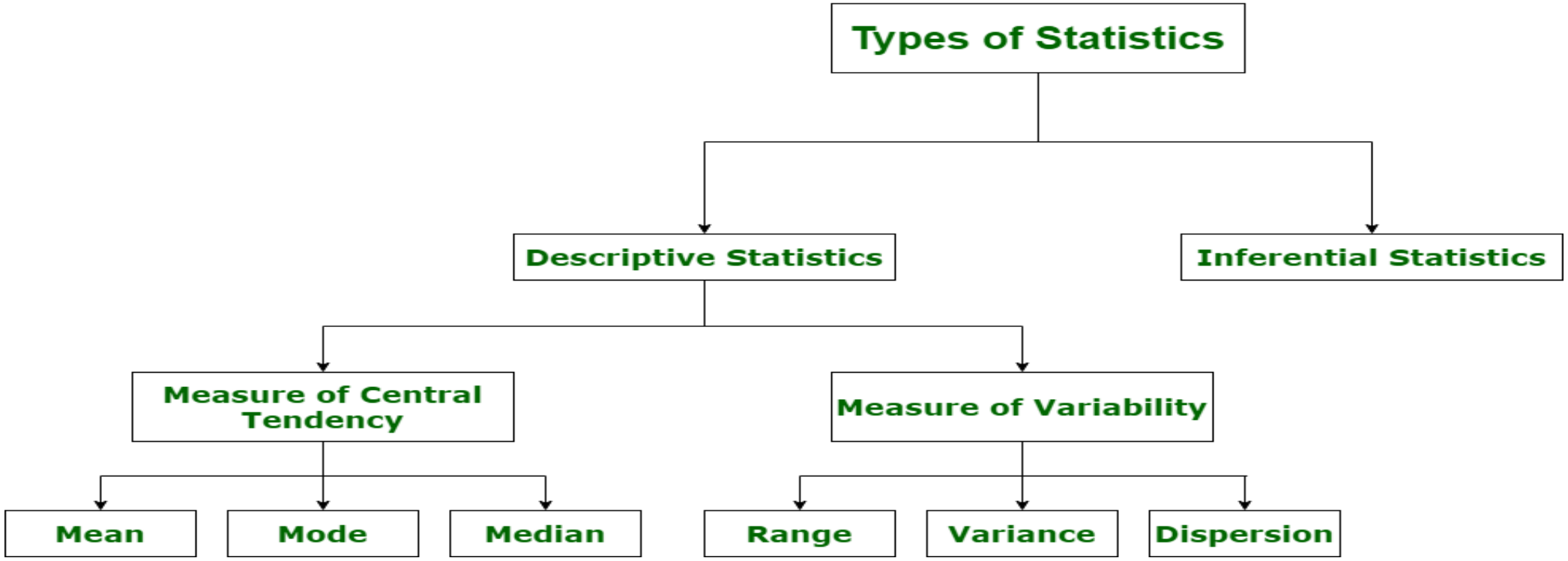
Attributes

- Attribute (or dimensions, features, variables)
  - A data field, representing a characteristic or feature of a data object.
  - E.g., customer_ID, name, address



Basic Statistical Descriptions of Data

- To better understand the data: central tendency, variation and spread



Data Visualization

Why data visualization?

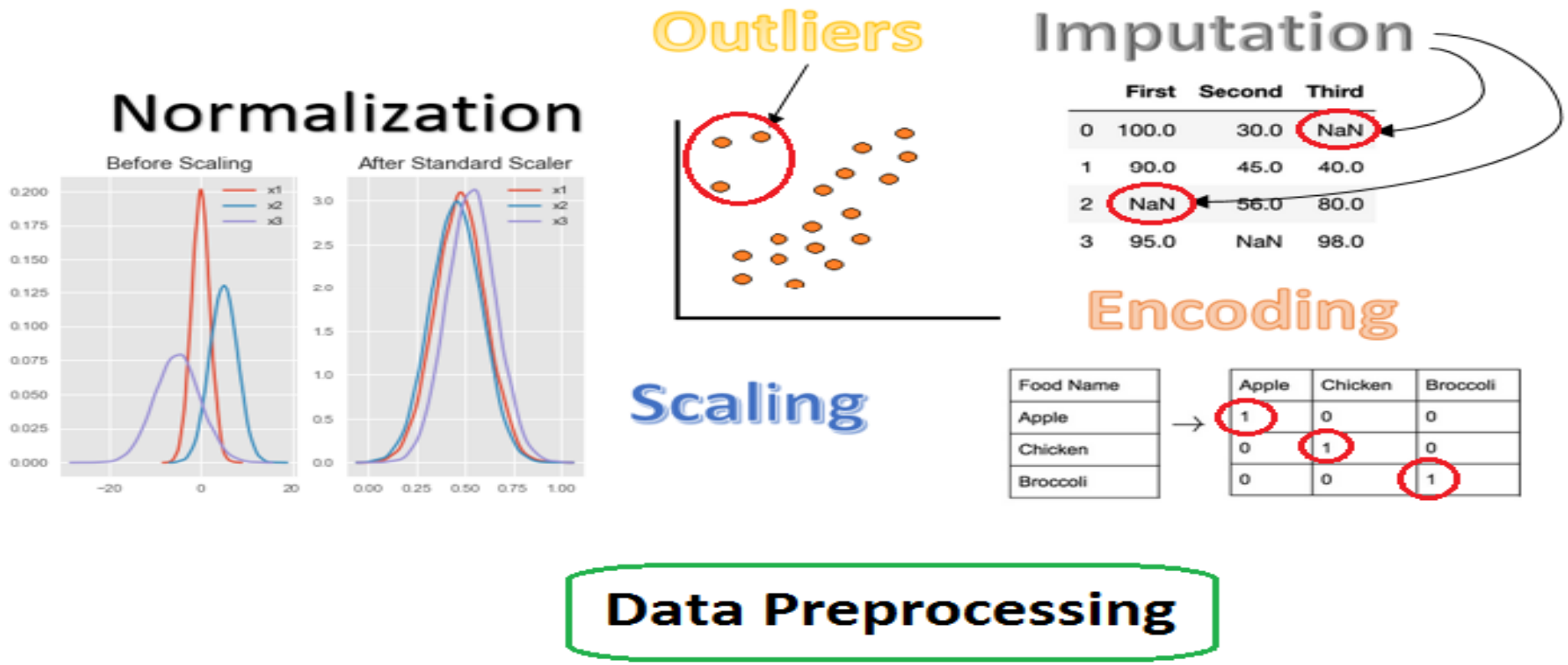
- Gain insight into an information space by mapping data onto graphical primitives
- Provide qualitative overview of large data sets
- Search for patterns, trends, structure, irregularities, relationships among data
- Help to find interesting regions and suitable parameters for further quantitative analysis

Data Mining

- Data Preprocessing
  - Data Cleaning
  - Data Integration
  - Data Reduction
  - Data Transformation and Data Discretization

Data Preprocessing

A robust method to make data ready for machine learning estimators.



Why Preprocess the Data?

- Real-world data are highly susceptible to
  - noisy, missing, and inconsistent data
- Reasons:
  - Huge size, multiple heterogenous sources, etc.
- Low-quality data → low-quality results

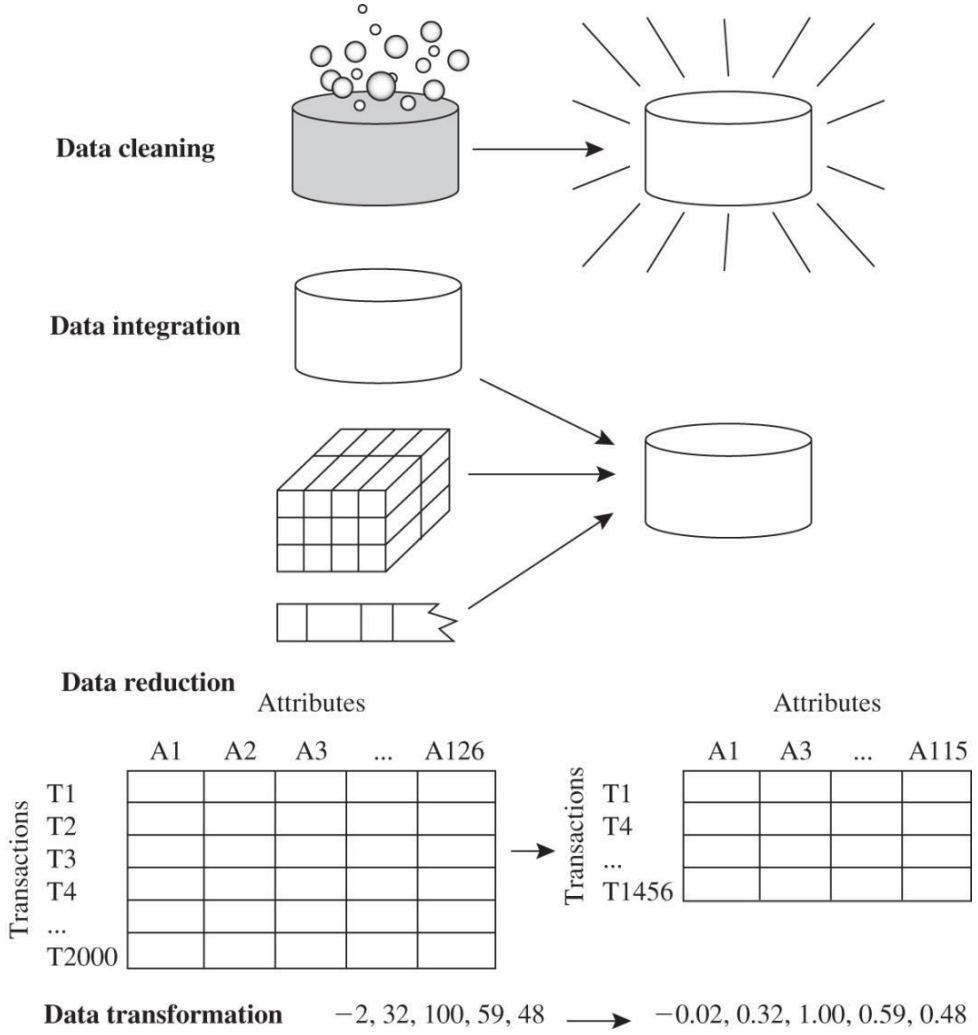
- Data preprocessing →
  - Improves the **quality of the data** → improves the results
  - Improves the efficiency

Data Quality Issues

- Measures for data quality
  - –Accuracy: correct or wrong, accurate or not
  - –Completeness: not recorded, unavailable, …
  - –Consistency: some modified but some not, …
  - –Timeliness: timely update?
  - –Believability: how trustable the data are correct?
  - –Interpretability: how easily the data can be understood?

Major Tasks in Data Preprocessing

- Data cleaning
  - –Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
  - –Integration of multiple data sources
- Data reduction
  - –Dimensionality reduction, numerosity reduction and data compression
- Data transformation and data discretization
  - –Normalization



Data Cleaning

- Data in the Real world is dirty: Lots of potentially incorrect data
  - E.g., instrument faulty, human or computer error, transmission error, etc.
- Data problems
  - Incomplete: lacking attribute values or lacking certain attributes
    - e.g., *Occupation*=“ ” (missing data)
  - Noisy: containing noise, errors, or outliers
    - e.g., *Salary*=“−10” (an error)
  - Inconsistent: containing discrepancies in codes or names, e.g.,
    - *Age*=“42”, *Birthday*=“03/07/2010”
    - Was rating “1, 2, 3”, now rating “A, B, C”
    - Discrepancy between duplicate records
  - Intentional (e.g., *disguised missing* data)
    - Jan. 1 as everyone’s birthday?

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (when doing classification)
  - not effective when the % of missing values per attribute varies considerably
- Fill in the missing value manually:
  - tedious + infeasible for large data size
- Fill in it automatically with
  - a global constant : e.g., “unknown” or “-∞”, a new class?!
  - a measure of central tendency for the attribute (e.g. attribute mean or median)
  - the attribute mean or median for all samples belonging to the same class: smarter
  - the most probable value: inference-based such as Bayesian formula or decision tree

Noisy Data

- Noise: random error or variance in a measured variable (Outliers)
- Detection
  - basic statistical description and data visualization techniques
- Incorrect attribute values may be due to
  - faulty data collection instruments
  - data entry problems
  - data transmission problems
  - technology limitation
- Other data problems which require data cleaning
  - duplicate records

How to Handle Noisy Data?

- Binning
  - first sort data and partition into (equal-frequency) bins
  - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Regression
  - smooth by fitting the data into regression functions

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:

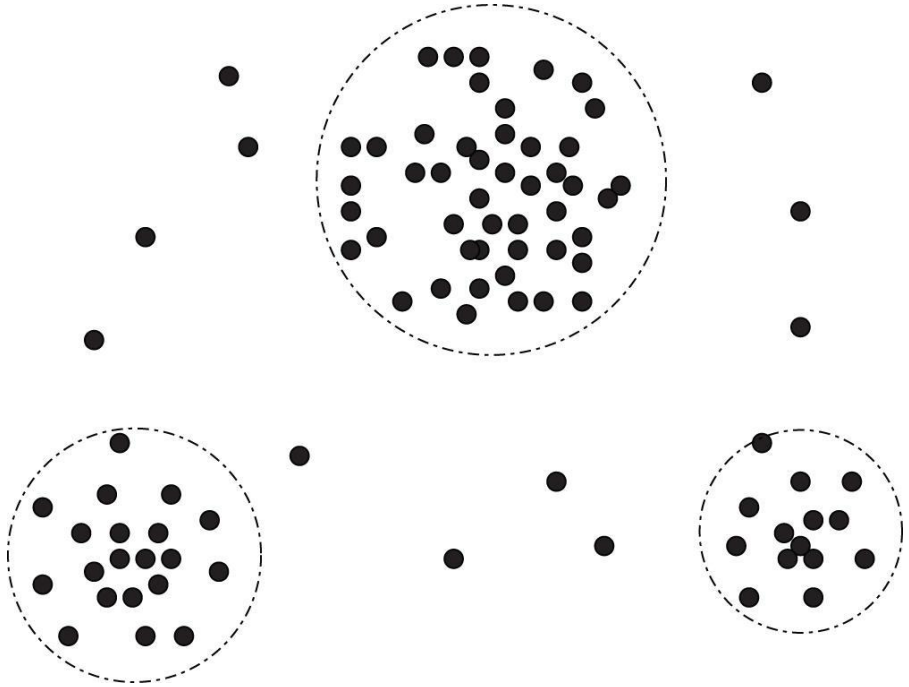
Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

How to Handle Noisy Data?

- Clustering
  – detect and remove outliers
- Combined computer and human inspection
  – detect suspicious values and check by human (e.g., deal with possible outliers)



Data Integration

- Data integration:
  - Combines data from multiple sources into a coherent store
- Schema integration: e.g., A.cust-id ≡ B.cust-#
  - Integrate metadata from different sources
- Detecting and resolving data value conflicts
  - For the same real world entity, attribute values from different sources are different
  - Possible reasons: different representations, different scales, e.g., metric vs. British units
- May result in data redundancy

Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
  - *Object identification*: The same attribute or object may have different names in different databases
  - *Derivable data:* One attribute may be a "derived" attribute in another table, e.g., annual revenue
- Redundant attributes may be able to be detected by ***correlation analysis*** and ***covariance analysis***
- Careful integration of the data from multiple sources may help to
  - Reduce/avoid redundancies and inconsistencies
  - Improve speed and quality

Dimensionality Reduction

- Data reduction:

Obtain a reduced representation of the data set that is **much smaller in volume but yet produces the same (or almost the same) analytical results**

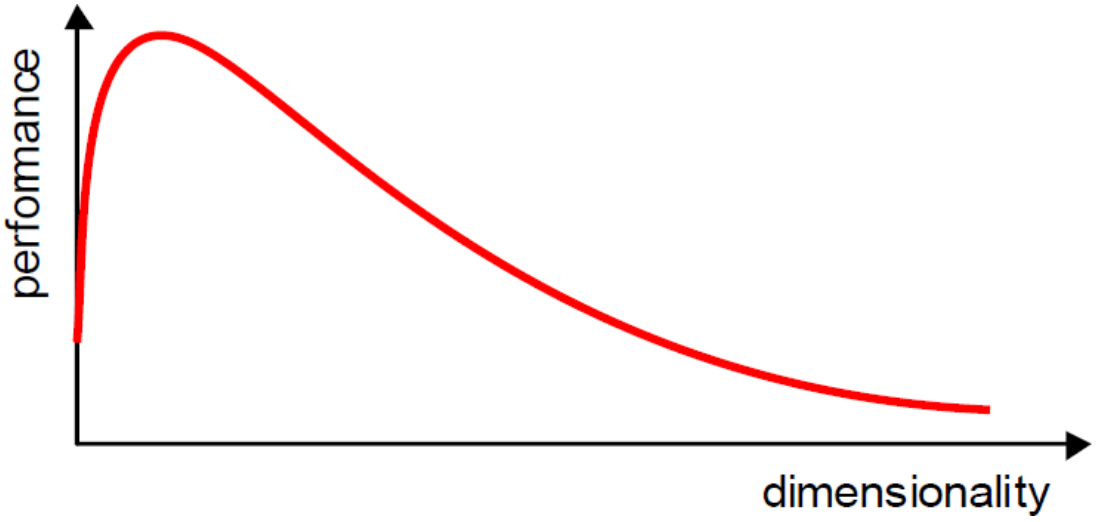
- Why data reduction?

A data set may contain terabytes of data → Complex data analysis may take a very long time to run on the complete data set.

- Increasing the number of inputs or features does not always improve accuracy of classification.
- Performance of classifier may degrade with the inclusion of irrelevant or redundant features.
- Curse of dimensionality:
  - When dimensionality increases, data becomes increasingly sparse
  - Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful

Data Reduction Strategies

1. Dimensionality reduction, e.g., remove unimportant attributes
   - Wavelet transforms
   - **Feature subset selection, feature creation or generation**
   - **Principal Components Analysis (PCA)**
2. Numerosity reduction (some simply call it: Data Reduction)
   - Regression and Log-Linear Models
   - Histograms, clustering, sampling
   - Aggregation
3. Data compression



Data Reduction Strategies

- Given a set of features, reduce the number of features such that "the learning ability of the model " is maximized. $\mathbf{x} = [x_1, x_2, \ldots, x_d]$

Feature Selection:

Select a subset of the existing features.

$$\mathbf{x} = [x_1, x_2, \ldots, x_d]$$

$$\Downarrow$$

$$\mathbf{z} = [x_{i_1}, x_{i_2}, \ldots, x_{i_k}]$$

Feature Extraction:

Transform existing features to obtain a set of new features using some mapping function.

$$\mathbf{x} = [x_1, x_2, \ldots, x_d]$$

$$\Downarrow \quad \mathbf{z} = f(\mathbf{x})$$

$$\mathbf{z} = [z_1, z_2, \ldots, z_k]$$

$$k \ll d$$

Feature Selection:

- Example:

$$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5] y$$

0	0	1	0	1	0
0	1	0	0	1	1
1	0	1	0	1	1
1	1	0	0	1	1
0	0	1	1	0	0
0	1	0	1	0	1
1	0	1	1	0	1
1	1	0	1	0	1

Data set:
- Five Boolean features
  - y= x_1 (or) x_2
  - x_3= (not) x_2
  - x_4= (not) x_5

Optimal subset:
{x1, x2} or {x1, x3}
Optimization in space of all feature subsets would have 2^d possibilities

Can't search over all possibilities and therefore we rely on heuristic methods

Heuristic Search in Attribute Selection

- How can we find a 'good' subset of the original attributes?

- There are 2^d possible attribute combinations of d attributes
  - An exhaustive search for the optimal subset of attributes can be prohibitively expensive

Typical heuristic attribute selection (greedy) methods (to reduce the search space):

- Best single attribute under the attribute independence assumption, chosen by:
  - Statistical significance tests
  - Information gain measure

1. Stepwise forward selection:
   - The best single-attribute is picked first
   - Then next best attribute condition to the first, ...

Forward selection	Backward elimination
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$
Initial reduced set: $\{\}$	=> $\{A_1, A_3, A_4, A_5, A_6\}$
=> $\{A_1\}$	=> $\{A_1, A_4, A_5, A_6\}$
=> $\{A_1, A_4\}$	=> Reduced attribute set: $\{A_1, A_4, A_6\}$
=> Reduced attribute set: $\{A_1, A_4, A_6\}$	

Data Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values
  - each old value can be identified with one of the new values
- Methods
  - Smoothing: Remove noise from data
  - Attribute/feature construction
    - New attributes constructed from the given ones
  - Aggregation: Summarization, data cube construction
  - Normalization: Scaled to fall within a smaller, specified range (e.g. [-1,1] or [0,1])
    - Min-max normalization
    - Z-score normalization
    - Normalization by decimal scaling
  - Discretization

Machine Learning

- What is Machine Learning
- Why Use Machine Learning
- Types of Machine Learning Systems
- Main Challenges of Machine Learning
- Machine Learning Cycle

What is Machine Learning

- Machine Learning is a subset of Data Mining focused on algorithms that learn patterns from data and generalize to unseen data.

- It gives computers the ability to learn without being explicitly programmed. (Arthur Samuel ,1959).

- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P,if its performance on T, as measured by P, improves with experience E.

What is Machine Learning

- ML focuses specifically on predictive modeling.

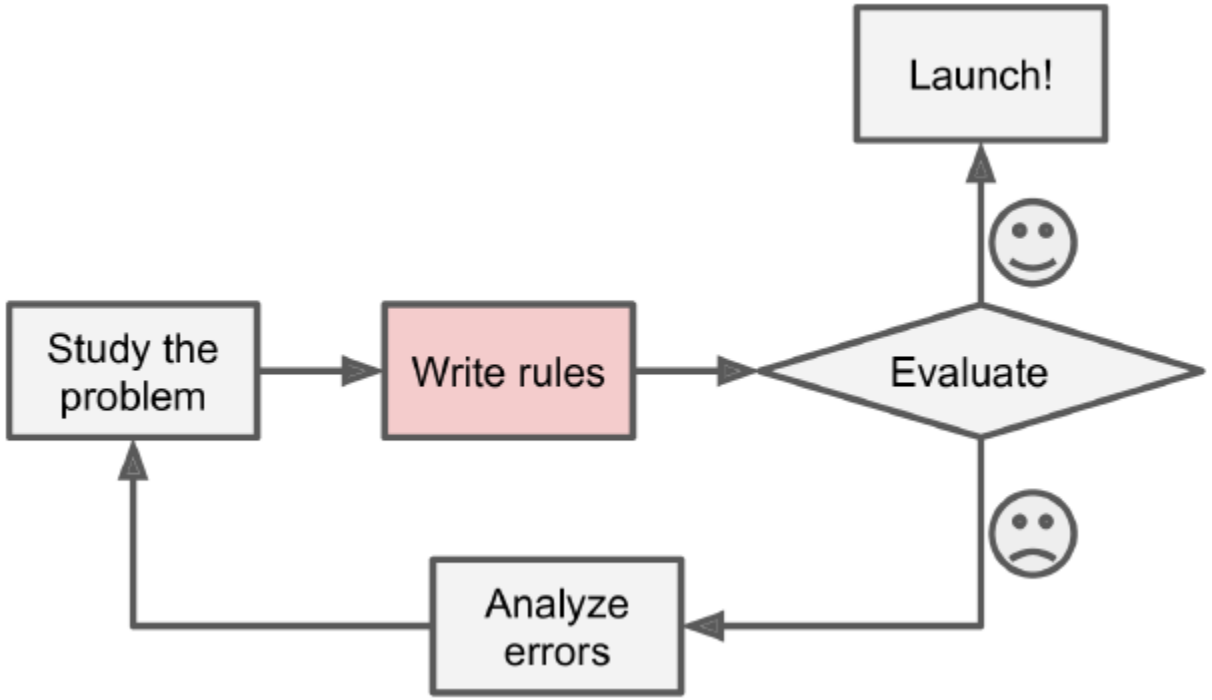
- Machine Learning:
  - Grew out of work in AI
  - New capability for computers
  - Data volume growth

Why Use Machine Learning?

- Take an example of spam filter

Traditional programming techniques:

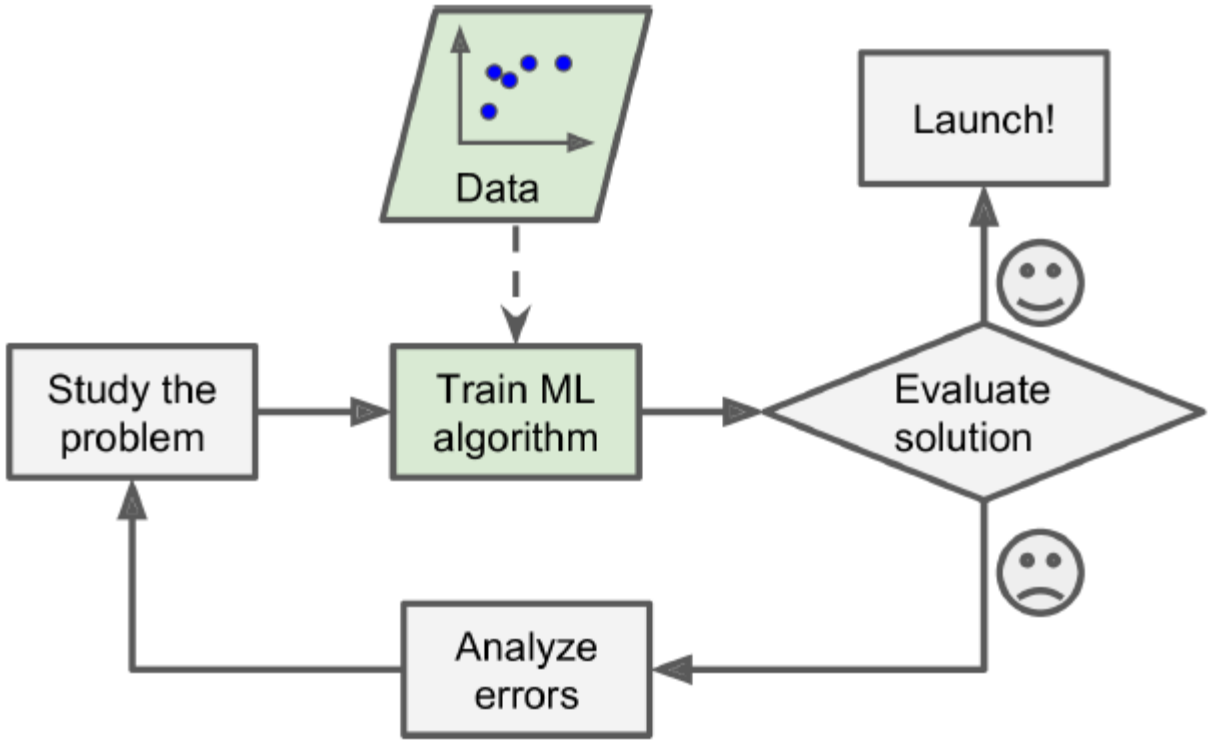
- Observe some words or phrases (“4U,” “credit card,” “free,” and “amazing”)
- –sender’s name, etc.
- Drawbacks:
  - Long list of complex rules
  - Hard to maintain



Why Use Machine Learning?

- ML Methods:
- Automatically learns which words and phrases are good predictors of spam
  - by detecting frequent patterns of words in the spam examples compared to the ham examples

- Advantages:
  - Shorter
  - Easier to maintain
  - More accurate



Machine Learning can be Used for:

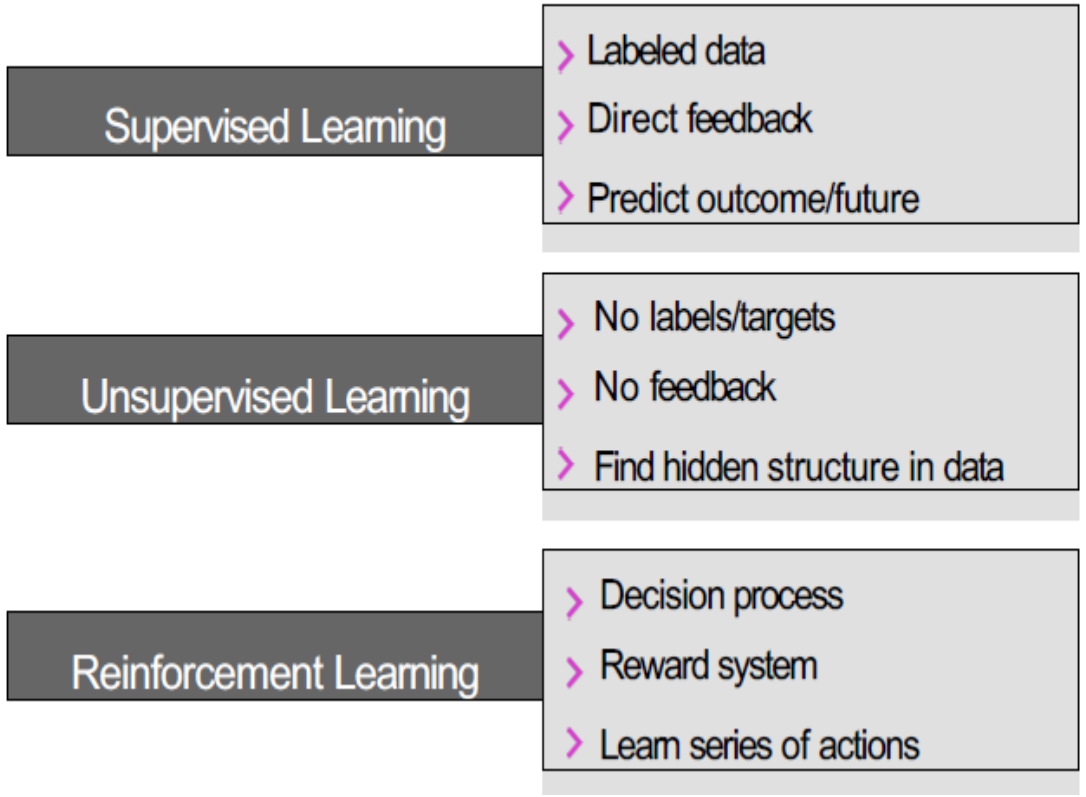
- Problems for which existing solutions require a lot of manual tuning or long lists of rules
  - Better performance
- Complex problems for which there is no good solution at all
- Fluctuating environments
  – Adaptive
- Getting insights about complex problems and large amounts of data
  – Discover unknown patterns

Types of Machine Learning Systems

- Trained with human supervision or not
  - supervised, unsupervised, semi-supervised, and Reinforcement Learning
- Can learn incrementally on the fly or not
  - Online (from a stream of incoming data) versus batch learning
- Working by comparing new data points to known data points, or detecting patterns in the training data and build a predictive model
  - instance-based versus model-based learning

Types of Machine Learning Tasks:

- ***Supervised Learning:***
  1. Regression (Continuous values)
  2. Classification (Categories)
- ***Unsupervised Learning:***
  1. Clustering
  2. Dimensionality reduction
  3. Anomaly detection
- Semi Supervised Learning
- Reinforcement Learning

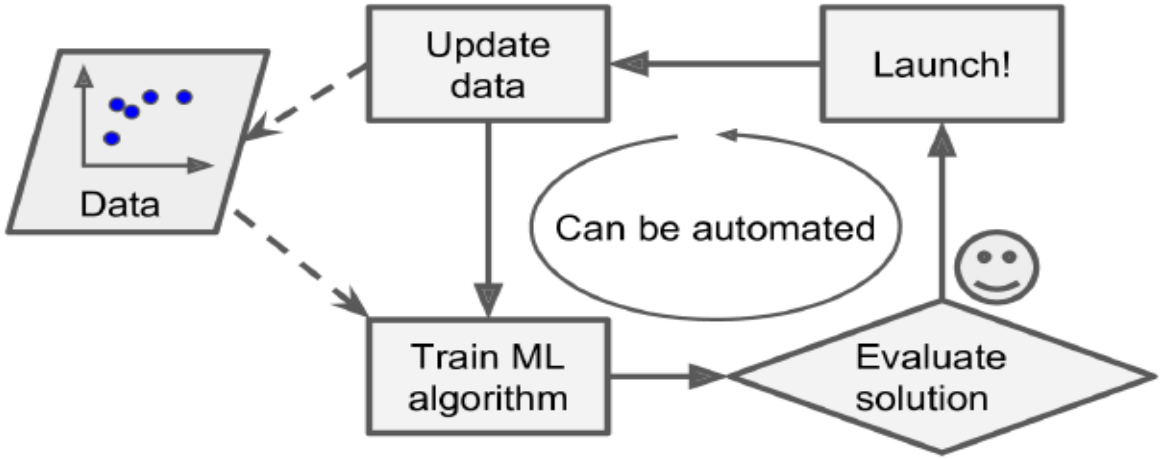


Batch Learning

- It must be trained using all the available data (Offline learning).

- Properties
  - Usually takes a lot of time (many hours) and computing resources
  - Runs without learning anymore (after finishing training)
  - New data → Train a new version from the beginning on the whole dataset
  - The whole process can be automated
- Not suitable for:
  - Rapidly changing data
  - Huge amount of data
  - Limited resources

Solution: Online learning



Online Learning

- The system is trained incrementally by providing data sequentially
  - Raw by raw or in mini-batches
- Properties
  - Learning process is fast and chap
  - Learning rate
    - High: rapidly adapt to new data
    - Low: less sensitive to noise

➢ Suitable for:

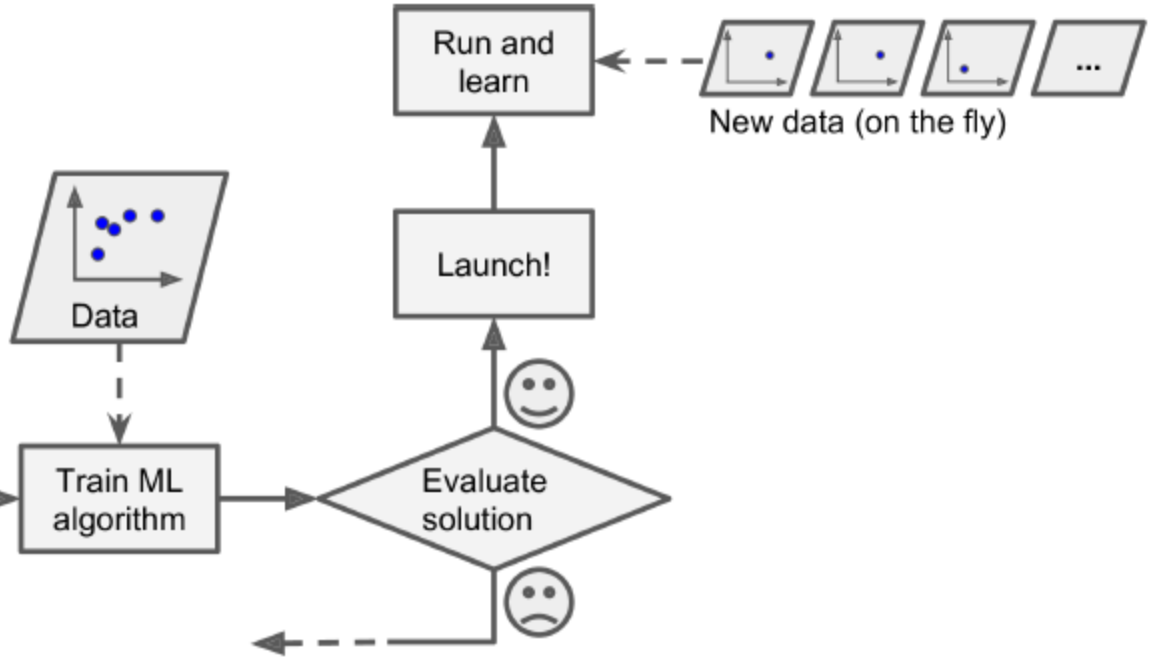
✓ Continuous data (e.g., stock prices)

✓ Rapidly changing data

✓ Autonomous learning

✓ Out-of-core learning (for huge data or limited resources)

  - Huge datasets that cannot fit machine's memory

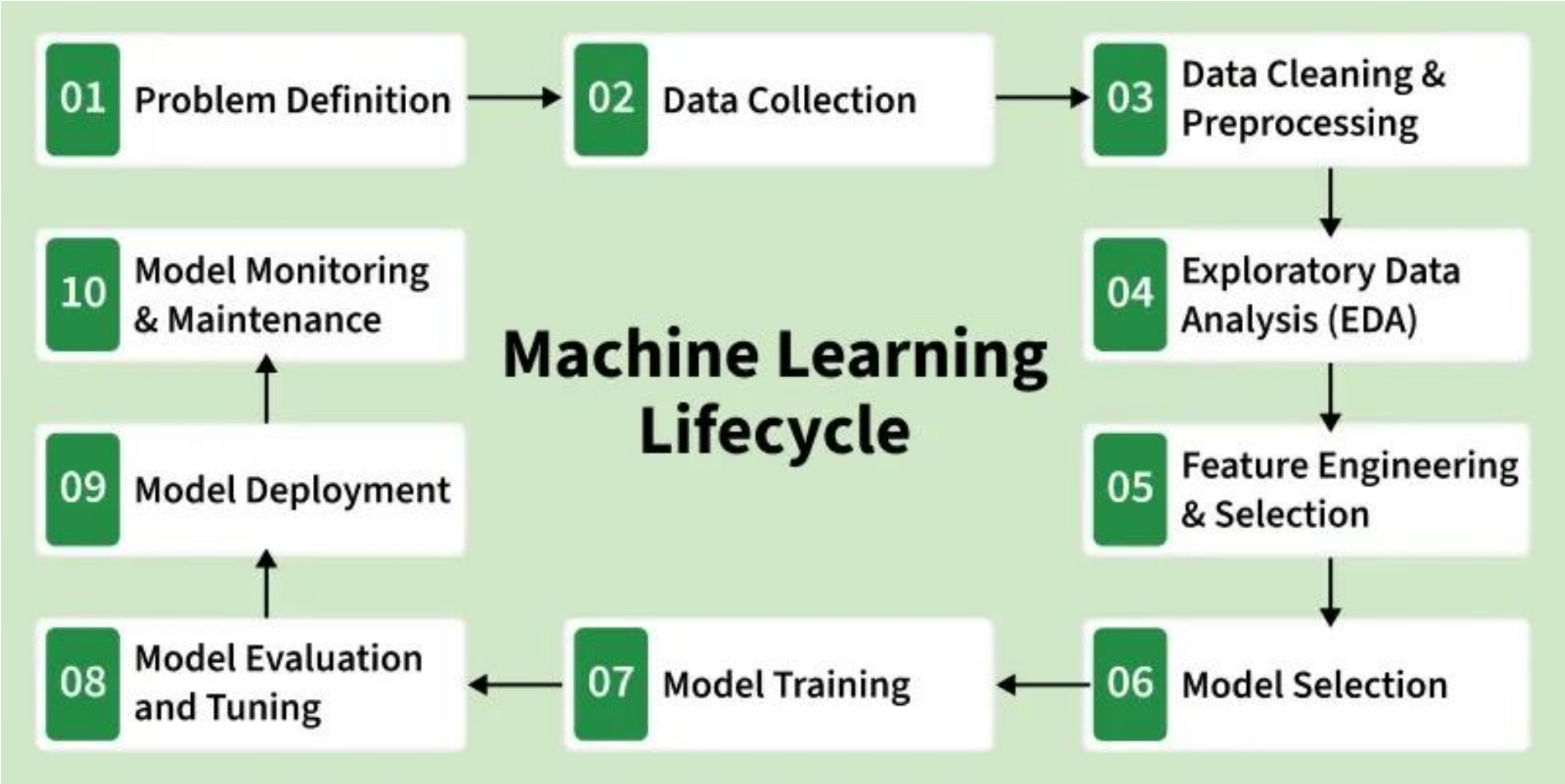


Main Challenges of Machine Learning

- Data Issues

1) Insufficient Quantity of Training Data
2) Non-representative Training Data
3) Poor-Quality Data
4) Irrelevant Features
5) Data Mismatch

- Algorithms

1) Overfitting the Training Data
2) Underfitting the Training Data

Machine Learning Cycle